RESEARCH + SOFTWARE

# ZHONGJIE HUANG

AI & Computer Science

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## PROFILE

I work across computer vision, machine learning evaluation and software development. My projects connect model experiments with usable systems, with particular attention to what the evidence supports and where it stops.

## EDUCATION

### University of Sheffield

2025-2026

MSc Artificial Intelligence

Dissertation submitted; final results and degree award pending.

### University of Sheffield

September 2021 - July 2025

BSc Artificial Intelligence and Computer Science

Awarded 1 July 2025 • Upper Second Class (2:1)

## SELECTED ACADEMIC PROJECTS

### Cell segmentation & downstream evidence

2026

Individual MSc dissertation project

- Compared QuPath, Cellpose-DINO, Cyto R-CNN and CSGO pipelines, examined their quantitative outputs, and assessed downstream models with attention to grouping and the independence of evaluation data.
- The dissertation did not establish a clear gain from fusion over the image-only approach. This makes the distinction between a better segmentation output, more useful measurements and better predictions central to the interpretation.

Python / QuPath / Cellpose-DINO / Cyto R-CNN / CSGO / Grouped evaluation

Project summary

### Emotion recognition using visual information

2025

Individual undergraduate project

- A complete image-to-prediction workflow, linking a Swin Transformer model with a responsive Java/Tomcat web application.
- The final report records 76.25% accuracy on its mixed test set.

Python / PyTorch / Swin Transformer / Java / Tomcat / HTML / CSS / JavaScript

Project summary

## PROPOSED DOCTORAL DIRECTION

### Reliable visual segmentation under environmental and sensor shift

A proposed doctoral direction: investigate whether confidence calibration, feature novelty and perturbation sensitivity help identify unreliable segmentation predictions when data conditions change. These are future research questions, not completed experiments.

## SELECTED SOFTWARE PROJECTS

### Marginloom - AI reliability workbench

2026

Personal research and software prototype, developed with AI assistance

- The repository implements temperature scaling, risk-coverage and threshold exploration, validated JSON prediction imports, saved run and policy snapshots, local ensemble inference, persisted review notes and JSON evidence exports. A separate companion experiment demonstrates segmentation on synthetic images.
- A React and TypeScript interface connects to typed API handlers and SQLite/D1 persistence. Python and scikit-learn produce reproducible baseline models and evaluation exports.

React / TypeScript / Python / scikit-learn / SQLite / D1 / Docker

Project summary | Public source

### IAFM - Investment & financial management

Portfolio project

Personal software project

- A Java application exploring structured financial records, validation, persistent storage and clear reporting interfaces.
- Layered Java design, persistence, input validation, access control and error handling, with separation between domain logic and presentation.

Java / SQL / Domain modelling / Access control / Dashboard UX

Project summary

## TECHNICAL TOOLKIT

**Programming & interfaces:** Python, Java, TypeScript, React, HTML / CSS / JavaScript

**Machine learning & vision:** PyTorch, Swin Transformer, OpenCV, QuPath, Cellpose-DINO, Cyto R-CNN, CSGO, scikit-learn

**Data & engineering:** NumPy, pandas, Matplotlib, SQL, SQLite / D1, Git, Docker, Linux, Tomcat, GitHub Actions

**Research practice:** Baselines, grouping and leakage checks, error analysis, traceable experiments, literature synthesis and clear statements of limitations.

**Communication:** Chinese; English - used in UK university study and technical communication